

Internal Troubleshooting Documentation

These articles represent internal support documentation designed to help agents resolve high-volume hardware and connectivity inquiries within a SaaS fitness platform.

Calibrating a Smart Trainer

Article Type: Troubleshooting

Product Area: Hardware Integration

Escalation Tier: Tier 3 if unresolved

Calibrating a smart trainer ensures accurate power readings and consistent resistance during workouts. Most issues stem from incomplete warmups, outdated firmware, connection conflicts, or unstable trainer setup.

Use this article when a user reports:

- Inconsistent resistance
- Power spikes or dropouts
- Failed spindown attempts
- "Calibration failed" errors

Pre-Calibration Checklist

Confirm the following before initiating calibration:

- Trainer is powered on and connected via Bluetooth or ANT+.
- Firmware is up to date.
- The app and companion apps are updated.
- No secondary devices or apps are connected to the trainer.

Calibration Steps

1. Have the user open the app's **Pairing / Connections** screen.
2. Confirm the trainer appears as both:
 - Power Source
 - Controllable Device
3. Select **Calibration / Spindown**.
4. Instruct the user to follow the on-screen prompts:
 - Accelerate to the target speed (typically 20–25 mph).
 - Stop pedaling when prompted.
 - Allow the flywheel to coast to a complete stop.
5. Wait for calibration confirmation (15–30 seconds).

If Calibration Fails

- Have the user complete a 10-minute warmup ride and retry.
- Confirm no other apps are connected via Bluetooth.
- Verify the trainer is on a stable surface with no wheel slippage.

If calibration continues to fail after these steps, escalate to Tier 3.

Escalation Criteria

Escalate if:

- Calibration repeatedly fails after warmup
- Firmware is current and pairing is correct
- No secondary connections are detected
- Hardware appears physically stable

Agent Notes

- Document firmware version before escalation.
 - Log ticket under: **Hardware → Trainer → Calibration Failure**.
 - Include device type, OS version, and connection method.
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Troubleshooting Heart Rate Monitor (HRM) Pairing

Article Type: Troubleshooting

Product Area: Device Connectivity

Escalation Tier: Tier 3 if unresolved

Pairing a heart rate monitor enables accurate BPM tracking during workouts. Connection failures are typically caused by battery issues, poor sensor contact, or Bluetooth conflicts.

Use this article when a user reports:

- HRM not appearing in pairing screen
- No BPM data displaying
- HRM disconnecting mid-session

Common Root Causes

- Low or dead battery
- Poor skin contact
- Multiple Bluetooth connections
- Outdated firmware
- Device interference

Initial Setup Verification

Confirm:

- HRM is powered on.
- Strap has direct skin contact.
- Conductive pads are lightly moistened (if applicable).
- Battery is fresh.
- Bluetooth or ANT+ is enabled.

Pairing Process

1. Have the user open the app's **Device Pairing** screen.

2. Select **Heart Rate**.
3. If multiple HRMs appear, confirm the correct model.
4. Verify live BPM data appears after pairing.

If No BPM Data Appears

- Unpair and re-pair the HRM.
- Instruct the user to restart the app.
- Confirm no other apps are connected to the HRM.

Additional Isolation Steps

- Have the user test the HRM in the manufacturer's app.
- If it fails there, advise contacting the manufacturer.
- If it works externally but not in-platform, escalate to Tier 3.

Escalation Criteria

Escalate if:

- HRM functions in external app but not in platform
- BPM drops persist after battery replacement
- Re-pairing and restart steps fail

Agent Notes

- Record HRM model and firmware (if available).
- Confirm device OS version.
- Log under: **Connectivity → HRM → Pairing Failure**.